Maths Explorers at the Terminal 11

Progressive exercises

E1 · Direct application. Calculate $\int_0^2 (3x^2 - 2x + 1)\ \mathrm{d}x$ by determining a antiderivative.

E2 · Application and justification. In an orthonormal coordinate system, calculate the area of the delimited domain by curves $y = x$ and $y = x^2$, between abscisses 0 and 1. Give the result in a unit of ,.

E3 · Method to be organised. Calculate the mean value of the exponential function over the interval $[0; 2]$.

E4 · Choice of strategy. Using part integration, calculate $\int_1^e \ln x\ \mathrm{d}x$.

E5 · Synthesis or demonstration. For every $n \in \mathbb{N}$, we're laying $I_n = \int_0^1 \frac{x^n}{1+x}\ \mathrm{d}x$, $x^0 = 1$ in Integrate. Demonstrate that $0 \leq I_n \leq \frac{1}{n+1}$, then deduct the limit of (I_n).